Logic, Language, and Philosophy

Proposition logic: natural deduction

Meeting 9, October 20, 2025

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Today

- Deductive systems
  - Validity and derivability
- Natural deduction
  - General characteristics
  - Conjunction
  - Implication
  - Repetition rule
  - Disjunction
  - To do
- Other approaches

Deductive systems

Deductive systems

Validity and derivability

Two perspectives

Two perspectives on the validity of an argument scheme $\phi_1 \ldots \phi_n / \psi$:

1. Semantic: given what $\phi_1 \ldots \phi_n$ and ψ *mean:*
   is the conclusion entailed by premisses?
2. Syntactic: given what $\phi_1 \ldots \phi_n$ and ψ *are:*
   can we derive the conclusion from the premisses?

Semantic view

Semantic view:

- proceeds in terms of meaning, and thus, extensionally, in terms of reference and truth
- so validity becomes: whenever the premisses are all true, the conclusion must be true as well

Notation:

$$\phi_1 \dots \phi_n \models \psi$$

Syntactic view

Syntactic view:

- proceeds in terms of form, thus, in terms of occurrences of logical constants
- so validity becomes: if the premisses are given, we can derive the conclusion

Notation:

$$\phi_1 \dots \phi_n \vdash \psi$$

Derivability

Deriving something is producing it from something else

- in a finite number of steps
- by application of rules that only refer to the form of something

Natural deduction

Natural deduction

General characteristics

Components

Two main components:

- Introduction rules: how do I get something of a certain form?
- Elimination rules: how can I use something of a certain form?

Derivations

A derivation of a formula ϕ is a finite, numbered and annotated list of formulas $\phi_1 \ldots \phi_m \ldots$ which has ϕ as its last element:

$$\begin{array}{ll} 1. & \phi_1 \\ \vdots & \\ m. & \phi_m \\ \vdots & \\ n & \phi \end{array}$$

Derivations (cont.)

Each row in a derivation is annotated to indicate how the formula on that row is obtained

Formulas can be obtained in two ways:

1. by application of one of a finite number of rules
2. by assumption

Indications:

1'. Rule and row number(s) of formula(s) on which the rule has been applied

2'. '*assumption*'

Derivations (cont.)

Thus:

$$
\begin{array}{lll}
1. & \phi_1 & \textit{assumption} \\
\vdots & & \\
m. & \phi_m & \textit{Rule}_k, j, l \\
\vdots & & \\
n & \phi & \textit{Rule}_l, h, m
\end{array}
$$

Assumptions

Assumptions can be:

a. Active
b. Withdrawn

Note:

- The premisses of an argument scheme are always active assumptions
- For reasons of perspicuity they are listed first in the derivation

Natural deduction

Conjunction

Conjunction introduction

Introduction rule $I_\wedge$:

$$
\begin{array}{lll}
1. & & \\
\vdots & & \\
m_1. & \phi & \\
\vdots & & \\
m_2. & \psi & \\
\vdots & & \\
n. & \phi \wedge \psi & I_\wedge, m_1, m_2
\end{array}
$$

Examples

$p, q, r \vdash q \wedge (p \wedge r)$

1.	p	A
2.	q	A
3.	r	A
4.	$p \wedge r$	$I_{\wedge}, 1, 3$
5.	$q \wedge (p \wedge r)$	$I_{\wedge}, 2, 4$

$p, q, r \vdash (p \wedge r) \wedge q$

1.	p	A
2.	q	A
3.	r	A
4.	$p \wedge r$	$I_{\wedge}, 1, 3$
5.	$(p \wedge r) \wedge q$	$I_{\wedge}, 2, 4$

Conjunction elimination

Elimination rule $E_\wedge$:

$$\begin{array}{lll} 1. & & \\ \vdots & & \\ m. & \phi \wedge \psi & \\ \vdots & & \\ k. & \phi & E_\wedge, m \end{array}$$

and:

$$\begin{array}{lll} 1. & & \\ \vdots & & \\ m. & \phi \wedge \psi & \\ \vdots & & \\ k. & \psi & E_\wedge, m \end{array}$$

Example

$p \wedge q \vdash q \wedge p$

1.	$p \wedge q$	A
2.	p	$E_\wedge, 1$
3.	q	$E_\wedge, 1$
4.	$q \wedge p$	$I_\wedge, 3, 2$

Natural deduction

Implication

Implication elimination

Elimination rule $E_{\rightarrow}$:

$$
\begin{array}{lll}
1. & & \\
\vdots & & \\
m_1. & \phi \rightarrow \psi & \\
\vdots & & \\
m_2. & \phi & \\
\vdots & & \\
k. & \psi & E_{\rightarrow}, m_1, m_2
\end{array}
$$

Example

$p \to (q \wedge r), r \to s, p \vdash s$

1.	$p \to (q \wedge r)$	A
2.	$r \to s$	A
3.	p	A
4.	$q \wedge r$	$E_{\to}, 1, 3$
5.	r	$E_{\wedge}, 4$
6.	s	$E_{\to}, 2, 5$

Implication introduction

Introduction rule $I_{\to}$:

$$\dfrac{\begin{array}{|lll} & 1. & \\ & \vdots & \\ & m. \quad \phi & \text{ass.} \\ & \vdots & \\ & n-1. \quad \psi & \end{array}}{n. \quad \phi \to \psi \quad I_{\to}}$$

Example

$p \to q, q \to r \vdash p \to r$

1.	$p \to q$	A
2.	$q \to r$	A
3.	p	A
4.	q	$E_\to, 1, 3$
5.	r	$E_\to, 2, 4$
6.	$p \to r$	$I_\to$

Assumptions

$$
\begin{array}{l}
\quad 1. \\
\quad \vdots \\
\left[\begin{array}{lll} m. & \phi & \text{ass.} \end{array}\right. \\
\quad \vdots \\
\left.\begin{array}{lll} n-1. & \psi & \end{array}\right] \\
\hline
\quad n. \quad \phi \to \psi \quad I_{\to}
\end{array}
$$

In this derivation the assumption ϕ:

- is made in step m
- is *withdrawn* in step n
- is *active* in steps $m+1$ to $n-1$

Conditions

Conditions:

1. An assumption that is withdrawn cannot be used later in the derivation
2. What depends on an assumption that is withdrawn also cannot be used later
3. When more than one assumption is made, the last one made is to be the first one that is withdrawn

Example condition 1

Condition 1:

1. An assumption that is withdrawn cannot be used later in the derivation

$$
\begin{array}{cccl}
\left\lceil\begin{array}{cccl} & 1. & p \wedge q & \mathtt{A} \\ & 2. & p & E_{\wedge}, 1 \end{array}\right. \\
\hline
& 3. & (p \wedge q) \to p & I_{\to} \\
* & 4. & q & E_{\wedge}, 1
\end{array}
$$

Example condition 2

Condition 2:

2. What depends on an assumption that is withdrawn also cannot be used later

$$
\begin{array}{cccl}
 & 1. & q \to r & \mathtt{A} \\
\left[\right. & 2. & p \wedge q & \mathtt{A} \\
\left.\right| & 3. & q & E_{\wedge}, 2 \\
\hline
 & 4. & (p \wedge q) \to q & I_{\to} \\
* & 5. & r & E_{\to}, 1, 3
\end{array}
$$

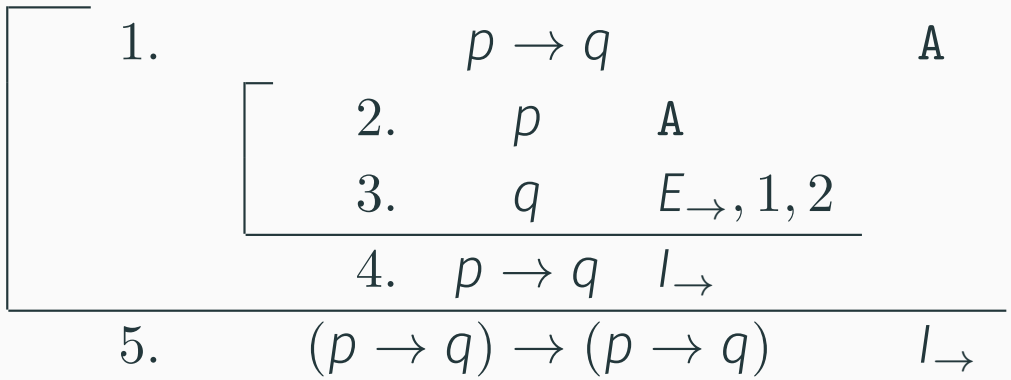
Example condition 3

Condition 3:

3. When more than one assumption is made, the last one made is to be the first one that is withdrawn

$$
\begin{array}{|lllll}
\hline
 & 1. & p \to q & \mathtt{A} & \\
 & 2. & p & \mathtt{A} & \\
 & 3. & q & E_{\to}, 1, 2 & \\
\hline
* & 4. & (p \to q) \to q & I_{\to} &
\end{array}
$$

Doing it right



Natural deduction

Repetition rule

Repetition

$$
\begin{array}{lll}
1. & & \\
\vdots & & \\
m. & \phi & \\
\vdots & & \\
n & \phi & \mathit{Rep}, m
\end{array}
$$

Condition:

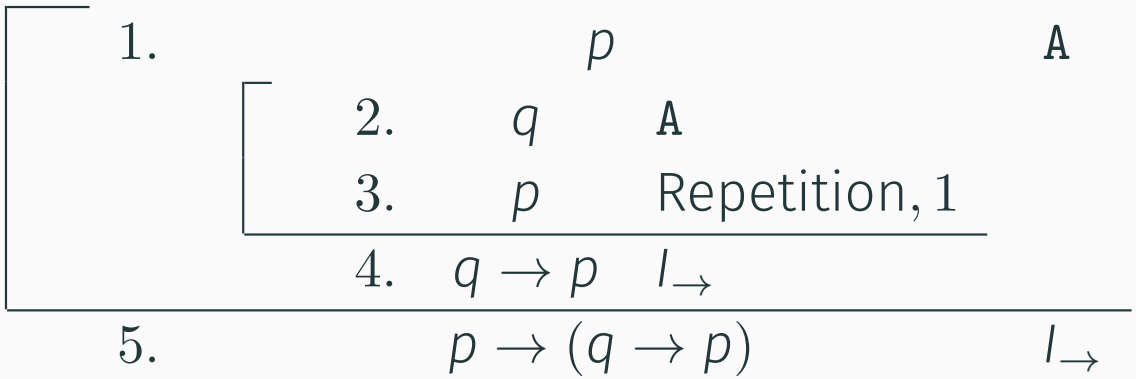
- The repeated formula is not an assumption that is withdrawn, or dependent on such an assumption

Note:

- Repetition does not add any expressive power to the system

Example

$\vdash p \to (q \to p)$:



Natural deduction

Disjunction

Disjunction introduction

Introduction rule $I_\vee$:

$$\begin{array}{lll} 1. & & \\ \vdots & & \\ m. & \phi & \\ \vdots & & \\ n. & \phi \vee \psi & I_\vee, m \end{array}$$

and:

$$\begin{array}{lll} 1. & & \\ \vdots & & \\ m. & \psi & \\ \vdots & & \\ n. & \phi \vee \psi & I_\vee, m \end{array}$$

Disjunction elimination

Elimination rule $E_\vee$:

$$
\begin{array}{lll}
1. & & \\
\vdots & & \\
m_1. & \phi \vee \psi & \\
\vdots & & \\
m_2. & \phi \to \chi & \\
\vdots & & \\
m_3. & \psi \to \chi & \\
\vdots & & \\
k. & \chi & E_\vee, m_1, m_2, m_3
\end{array}
$$

Example

$p \vee q \vdash ((p \to q) \to q)$:

1.	$p \vee q$		A
2.	p		A
3.	$p \to q$	A	
4.	q	$E_{\to}, 2, 3$	
5.	$(p \to q) \to q$	$I_{\to}$	
6.	$p \to ((p \to q) \to q)$		$I_{\to}$
7.	q		A
8.	$p \to q$	A	
9.	q	Repetition, 7	
10.	$(p \to q) \to q$	$I_{\to}$	
11.	$q \to ((p \to q) \to q)$		$I_{\to}$
12.	$(p \to q) \to q)$		$E_{\vee}, 1, 6, 11$

Natural deduction

To do

Still to do

- Negation

Other approaches

Approaches

Besides natural deduction systems there are:

- Axiomatic systems
- Sequent calculi
- Resolution logics
- ...

Hilbert-style axiomatisation: example

Łukaciewicz' axiomatisation of proposition logic:

- Axioms:
  1. $\phi \to (\psi \to \phi)$
  2. $(\phi \to (\psi \to \chi)) \to ((\phi \to \psi) \to (\phi \to \chi))$
  3. $(\neg\phi \to \neg\psi) \to (\psi \to \phi)$
- Rule of inference:
  1. Modus Ponens: from ϕ and $\phi \to \psi$, conclude ψ

Example

$\vdash p \to p$

1. $p \to (p \to p)$ (A1, by $[p/\phi][p/\psi]$)
2. $p \to ((p \to p) \to p)$ (A1, by $[p/\phi][p \to p/\psi]$)
3. $(p \to ((p \to p) \to p)) \to ((p \to (p \to p)) \to (p \to p))$ (A2, $[p/\phi][p \to p/\psi][p/\chi]$)
4. $(p \to (p \to p)) \to (p \to p)$ (MP 2, 3)
5. $p \to p$ (MP 1, 4)

Axioms versus natural deduction

- An axiom system and a natural deduction system capture the same concept: derivability in standard propositional logic
- But they do this in a different manner:
  - A natural deduction system has rules for each connective that tell us how it can be used
  - An axiom system does not tell us how to construct a derivation
- Also: the axioms of an axiom system (often) lack intuitiveness
- A natural deduction system captures a particular philosophical outlook on meaning, i.e., that meaning as use

Homework

Make exercises 3, 4, 5, 6 and 7.